

Age & Blood Pressure Regression Analysis

A Simple Linear Regression & Correlation Study

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Course: Statistics II

DATASET

NAME	A	B	C	D	E	F	G	H	I	J
AGE(x)	42	36	55	58	35	65	60	50	48	51
BP(y)	98	93	110	85	105	108	82	102	118	99

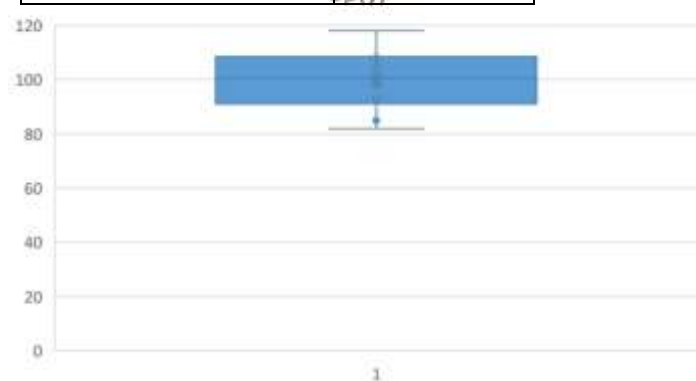
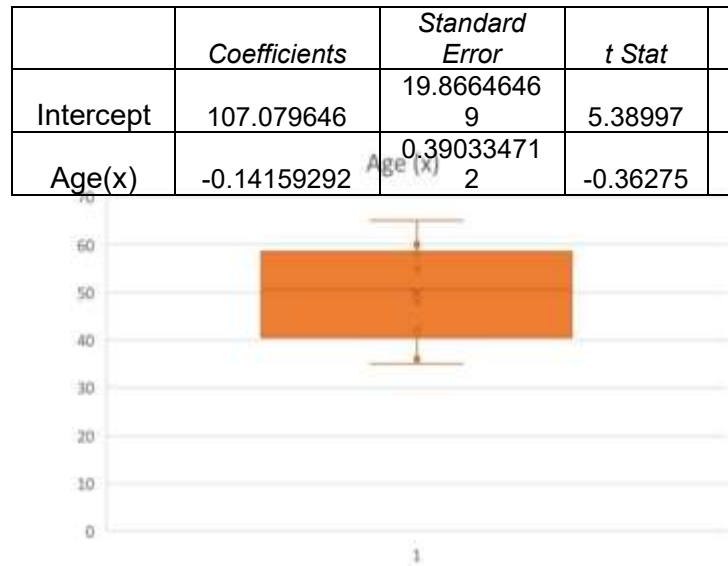
Dataset Overview

The dataset presents the age (x) and corresponding blood pressure (BP) (y) readings of 10 individuals labeled A to J. Age (x) serves as the independent variable, while BP (y) is the dependent variable. This data is suitable for analyzing potential relationships between age and blood pressure, such as correlation or regression modeling.

DESCRIPTIVE STATISTICS

Age(x)	
Mean	50
Standard Error	3.40116175
Median	50.5
Standard Deviation	10.0222
Sample Variance	100.4444
Kurtosis	1.939512
Skewness	-0.1891768
Range	30
Minimum	35
Maximum	65
Sum	500
Count	10

						Bp(y)		
						Mean	100	
						Standard Error	3.936219908	
						Median	100.5	
						Standard Deviation	11.15547	
						Lower 95.0%	82.44445	
						Upper 95.0%	118.55555	
						Skewness	-0.17565	
						Kurtosis	0.75852	
						Range	36	
						Minimum	82	
						Maximum	118	
						Sum	1000	
						Count	10	
						Lower 95.0%	61.26749	
						Upper 95.0%	152.89179	
						Skewness	-0.17565	
						Kurtosis	0.75852	
						Range	36	
						Minimum	82	
						Maximum	118	
						Sum	1000	
						Count	10	



Regression Analysis

Multiple R	0.127208679
R Square	0.016182048
Standard Error	11.73603482
Observations	10

Coefficient of correlation (r) = 0.12727

Coefficient of determination (r^2) = 0.0162

The coefficient of correlation (r) is 0.12727, which indicates a **very weak positive linear relationship** between Age and Blood Pressure. This means that as age increases, blood pressure tends to increase slightly, but the relationship is **not strong or significant**.

$$\hat{Y} = a + bx$$

$$\hat{Y} = 107.079646 - 0.14159292X$$

Estimated Values

<i>Observation</i>	<i>Predicted AGE(X)</i>
1	50.22857143
2	50.8
3	48.85714286
4	51.71428571
5	49.42857143
6	49.08571429
7	52.05714286
8	49.77142857
9	47.94285714
10	50.11428571

Confidence and Prediction Interval

90% confidence interval and prediction interval for age and BP levels of patients at age 36.

$$\hat{Y}=101.9824$$

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	14.62857143	14.6285714	0.13158571	0.726190425
Residual	8	889.3714286	111.171429		
Total	9	904			

$$n=10$$

Standard Error of Estimate:

$$\sqrt{[\sum(Y - \hat{Y})^2 / (n - 2)]}$$

$$S_{y.x}=11.832$$

$$t=1.860$$

Confidence Interval:

$$\hat{Y} \pm t (S_{y.x}) \sqrt{1/n + (X - \bar{X})^2 / \sum ((X - \bar{X})^2)}$$

$$CI: 101.9824 \pm 1.860(11.736) \sqrt{0.5628}$$

CI is between **89.695** and **114.2724**

Prediction Interval:

$$\hat{Y} \pm t (S_{y.x}) \sqrt{1 + 1/n + (X - \bar{X})^2 / \sum ((X - \bar{X})^2)}$$

$$PI: 101.9824 \pm 1.860(11.736) \sqrt{1.5628}$$

PI is between **67.8668** and **136.098**

Anova Table

Hypothesis Tests

Hypothesis test for correlation:

$$n=10$$

$$r=0.12727$$

$$\alpha=0.05$$

$$H_0: \rho \leq 0$$

$$H_1: \rho > 0$$

$$t_{cal} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \quad t_{cal} = 0.363$$

$$t_{\alpha, (n-2)} = 1.860$$

Reject H_0 if $t_{cal} > t_{\alpha, (n-2)}$

$$0.363 \ngtr 1.860$$

$\therefore$ Do not reject H_0

Hypothesis test for slope:

$$\hat{Y} = 107.079646 - 0.14159292X$$

$$\alpha = 0.05$$

$$S_b = 0.3903$$

$$H_0: \beta \geq 0$$

$$H_1: \beta < 0$$

$$t = \frac{b - 0}{S_b}$$

$$t = -0.363$$

$$t_{\alpha, (n-2)} = 1.860$$

Reject H_0 if $t_{cal} > t_{\alpha, (n-2)}$

$$-0.363 > -1.860$$

$\therefore$ Reject H_0

Interpretation: We can say that there is no significant relationship between the slopes of both the variables.

